

FSM DIAGRAMS

PART A:

STATES IN THE DIAGRAM:

A → Initial state or state in which the last four inputs are in the form XX00

B → State in which the last four inputs are of the form X001 or XX11

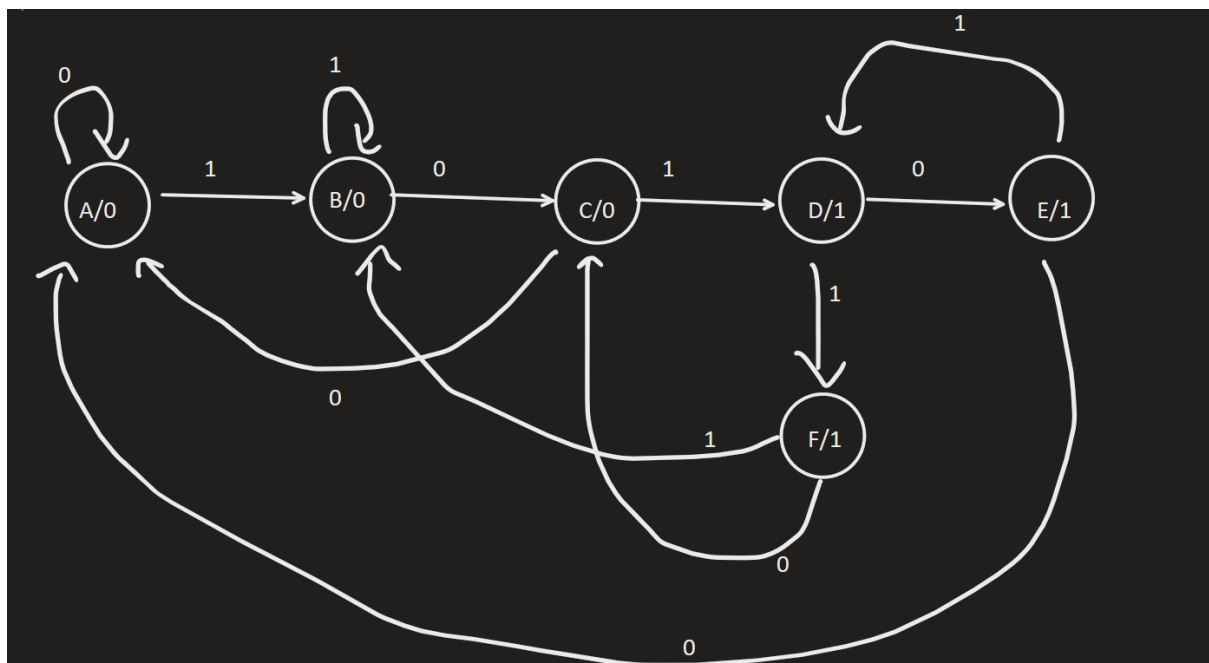
C → State in which the last four inputs are of the form XX10 but not 1010

D → State in which the last four inputs are of the form X101

E → State in which the last four inputs are 1010

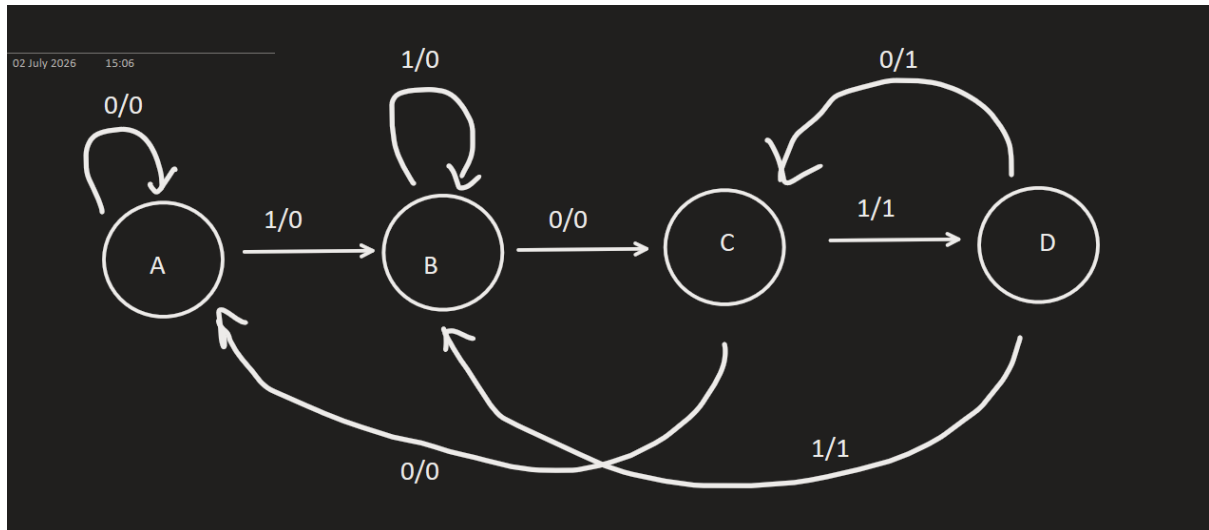
F → State in which the last four inputs are 1011

MOORE FSM DIAGRAM



MEALY FSM DIAGRAM:

the numbers on the arrows in the mealy FSM diagram are in the form input/output i.e. 0/0 means the input given is 0 and the output received is 0;



PART B:

FSM FRAME RECEIVER:

This is implemented through a MEALY FSM.

STATES IN THE DIAGRAM:

0 → initial state

1 ... 8 → states in which the input is stored in data_out_temp_store register, and the parity is calculated, if done == 1 then the value stored in the data_out_temp_store is given to data_out as output during state 10.

9 → state in which parity as is taken as input. If the input is equal to the calculated parity in states 1....8 then parity_err is zero else it is 1.

10 → state in which stop bit is taken as input. If input is one and no errors exist then value of data_out_temp_store is given to data_out and done is set to 1. Else frame_err is set to 1.

